BJA

British Journal of Anaesthesia, 134 (3): 736–745 (2025)

doi: 10.1016/j.bja.2024.10.051
Advance Access Publication Date: 23 January 2025
Clinical Investigation

NEUROSCIENCE AND NEUROANAESTHESIA

Qnox index for quantification of intraoperative nociception and analgesia: a prospective single-centre validation study

Hao Kong[1], Dan-Dan Ma[2], Jia-Hui Ma[1], Yu-Xiu Zhang[1], Hong Zhang[1,*] and Dong-Xin Wang[1,3]

[1]Department of Anaesthesiology, Peking University First Hospital, Beijing, China, [2]Department of Anaesthesiology, Yidu Central Hospital, Qingzhou, China and [3]Outcomes Research Consortium, Cleveland, OH, USA

*Corresponding author. E-mail: zhanghong40@hotmail.com

Abstract

Background: The Qnox index is a novel monitor to quantify intraoperative nociception based on an electroencephalographic algorithm. We evaluated the ability of the Qnox index to discriminate noxious from non-noxious stimuli, respond to stimuli, and discriminate different levels of analgesia in patients under propofol anaesthesia with neuromuscular block.

Methods: Qnox was compared with heart rate and mean arterial pressure with five designated stimuli: tetanic stimulations without (tetanic 1) and with sufentanil (tetanic 2), skin incision, tracheal intubation, and a non-noxious period. The response around the skin incision was also evaluated at two target remifentanil concentrations.

Results: In 83 adult patients scheduled for elective surgery, Qnox performed worse than heart rate and mean arterial pressure in discriminating tetanic 2, tetanic 1, skin incision, and tracheal intubation noxious stimuli from the non-noxious period, with an area under curve of 0.52 (95% confidence interval 0.43–0.61), 0.54 (0.45–0.62), 0.67 (0.58–0.75), and 0.65 (0.57–0.73), respectively. The post-stimulus values of Qnox increased significantly after tracheal intubation and skin incision, but not after tetanic 1 or tetanic 2. Qnox values after skin incision were similar between the low- and high-remifentanil-concentration groups.

Conclusions: Qnox had a poor ability to discriminate noxious stimuli from non-noxious stimuli. Although Qnox responded to tracheal intubation and skin incision, it did not respond to tetanic stimulations and failed to discriminate different levels of analgesia. The Qnox index was not superior to heart rate or mean arterial pressure in assessing nociception during general anaesthesia.

Clinical trial registration: Chinese Clinical Trial Registry (ChiCTR2100046063).

Keywords: analgesia; heart rate; intraoperative nociception; mean arterial pressure; noxious stimuli; Qnox index

Editor's key points

- The Qnox index is an EEG-based monitor to quantify intraoperative nociception but its clinical utility is unclear.
- The ability of the Qnox index to discriminate noxious from non-noxious stimuli under propofol anaesthesia with neuromuscular block was tested compared with heart rate and mean arterial pressure in adults undergoing elective surgery.
- Qnox was not superior to heart rate and mean arterial pressure in discriminating noxious stimuli from a non-noxious period.
- Although Qnox responded to tracheal intubation and skin incision, it failed to discriminate different levels of remifentanil analgesia during propofol anaesthesia.

Received: 20 May 2024; **Accepted:** 18 October 2024

© 2025 British Journal of Anaesthesia. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.
For Permissions, please email: permissions@elsevier.com

Nociception, in contrast to pain, is not a subjective experience, but rather a physiological term to describe the encoding and processing of stimuli by the autonomic nervous system.[1] Nociceptive signals are continuously generated during surgery, even in the presence of deep anaesthesia. Intraoperative nociception is associated with adverse outcomes, triggering haemodynamic fluctuations, reflex movements, and excessive stress response.[1] Therefore, it is important to identify and measure the balance of nociception and antinociception intraoperatively. However, the gold standard for intraoperative nociception monitoring with accuracy, convenience, and easy interpretation is still not established[2] as all commercially available monitors have shown limitations.[1,2]

The CONOX Monitor (Quantium Medical S.L., Mataro, Spain) provides a 0–99 dimensionless score based on the analysis of the raw electroencephalogram (EEG) using a mathematical model: the adaptive neuro-fuzzy inference system (ANFIS).[3] The monitor includes two indices, the Qcon index for hypnotic level and the Qnox index for nociception. The index was developed with an open algorithm using the higher frequencies of the EEG power spectrum for nociception, with somewhat lower frequencies used for level of consciousness.[3,4] According to the manufacturer, a Qnox value below 40 signifies a very low likelihood, 40–60 a low likelihood, and more than 60 a higher likelihood of a response to a noxious stimulus.[1] In the conscious state, the values of Qnox and Qcon are above 90. Values of Qcon and Qnox between 40 and 60 are considered appropriate for intraoperative levels of anaesthesia and nociception.[5] The proprietary index is also known in the Asiatic market as IOC1 and IOC2 (Angel-6000 Multi-parameter Anaesthesia Monitor; Shenzhen Weihaokang Medical Technology Co., Ltd, Guangdong, China).[6]

Unlike other nociception monitors that rely almost entirely on the assessment of autonomic tone, Qnox is independent of autonomic activity, which might render the score more robust against the influence of arrhythmias and vasoactive drugs.[2] Although existing evidence is inadequate to determine the clinical validity of the score definitively, the appealing features of the index make it a promising option for nociceptive assessment.

Jensen and colleagues,[3] the inventors of the algorithm, and Melia and colleagues[7] claimed that Qnox could better predict movement in response to severe noxious stimuli such as laryngeal mask airway insertion or tracheal intubation; however, whether Qnox has the ability to respond to mild or moderate stimuli was inconclusive. In addition, the effect of the electromyogram (EMG) on the Qnox index was not confirmed because neuromuscular blocking agents were not used in these studies.[3,7] Rogozov and colleagues[8] observed that the Qnox index showed a specific relationship with the EMG, and simultaneous muscle electrical activity might influence detection or processing. To further verify the accuracy of Qnox, we evaluated the abilities of Qnox to differentiate mild, moderate, and severe levels of stimulation from non-noxious stimuli, respond to graded noxious stimuli, and discriminate different levels of analgesia under general anaesthesia with neuromuscular blocking agents.

Methods

The study protocol was approved by the Biomedical Research Ethics Committee of Peking University First Hospital (No. 2021-004; February 10, 2021) and registered in the Chinese Clinical Trial Registry (No. ChiCTR2100046063; May 3, 2021). The study was initiated on May 21, 2021 and concluded on January 31, 2023. We obtained written informed consent from all subjects.

Participants

Potential participants were screened on the day before surgery. We included adult patients (aged 18 yr or more) who were scheduled for elective surgery under general anaesthesia. Patients who met any of the following criteria were excluded: (1) cognitive impairment; (2) pregnant or lactating; (3) mental illness or nervous system disease; (4) abuse of alcohol or illegal drugs; or (5) severe cardiac arrhythmia.

Anaesthesia and stimulation

Patients received standard anaesthesia monitoring, including electrocardiogram, blood pressure (BP), heart rate (HR), pulse oxygen saturation, capnography, and nasopharyngeal temperature. An intra-arterial catheter was inserted to monitor beat-to-beat arterial pressure. Bispectral index (BIS; Medtronic, Minneapolis, MN, USA) was used to monitor hypnotic depth. Qcon index, which assessed the level of consciousness, and the Qnox index, which assessed the level of noxious stimulation during surgery, were monitored with the Angel-6000D Multi-parameter Anaesthesia Monitor (Shenzhen Weihaokang Medical Technology Co., Ltd).

Tetanic stimulation electrodes were placed over the ulnar nerve on the ulnar side of the wrist. Tetanic stimulus before analgesic administration (tetanic 1), tetanic stimulus after analgesic administration (tetanic 2), tracheal intubation, and skin incision were designated as four noxious stimuli. The intensity of stimulation was *a priori* ordered based on previous publications[5,9] as tracheal intubation (moderate to severe) > skin incision (moderate) > tetanic 1 (mild-to-moderate) > tetanic 2 (mild) > non-noxious period (no pain).

Anaesthesia was induced and maintained with midazolam 0.02 mg kg^{-1} and target-controlled infusion (TCI) of propofol at 1 μg ml^{-1} increments; the target was to maintain BIS between 40 and 60, after which a dose of rocuronium 0.8 mg kg^{-1} was administered. Before tracheal intubation, patients were subjected to tetanic stimulation (50 mA, 100 Hz, 20 s) without analgesic administration (tetanic 1) and the same intensity of tetanic stimulation after a bolus of 0.2 μg kg^{-1} sufentanil (tetanic 2), with an interval of 3 min. After tracheal intubation, general anaesthesia was maintained with adjustments of propofol concentration to a BIS target between 40 and 60. Then, position was adjusted to ensure safety and facilitate access to the surgical site. Before skin incision, participants were randomly allocated to remifentanil target plasma concentrations of 2 or 4 ng ml^{-1}. Randomisation to remifentanil concentration groups, with a ratio of 1:1, was done using the SAS statistical package version 9.3 (SAS Institute, Cary, NC, USA) by an independent biostatistician. An independent anaesthesia assistant who was otherwise not involved in the study set remifentanil concentrations based on the results of random envelopes and covered the TCI pump screen with white cardboard. The assistant ensured the effect-site concentration had reached the predetermined concentration before skin incision. Anaesthesiologists and participants were blinded to the assigned group throughout the study. A 5-min period without pain and any noxious stimulus was designated as a non-noxious period before the initial skin incision. Figure 1 illustrates the timeline of the study protocol. Rescue sufentanil at a 5-μg dose was allowed if the

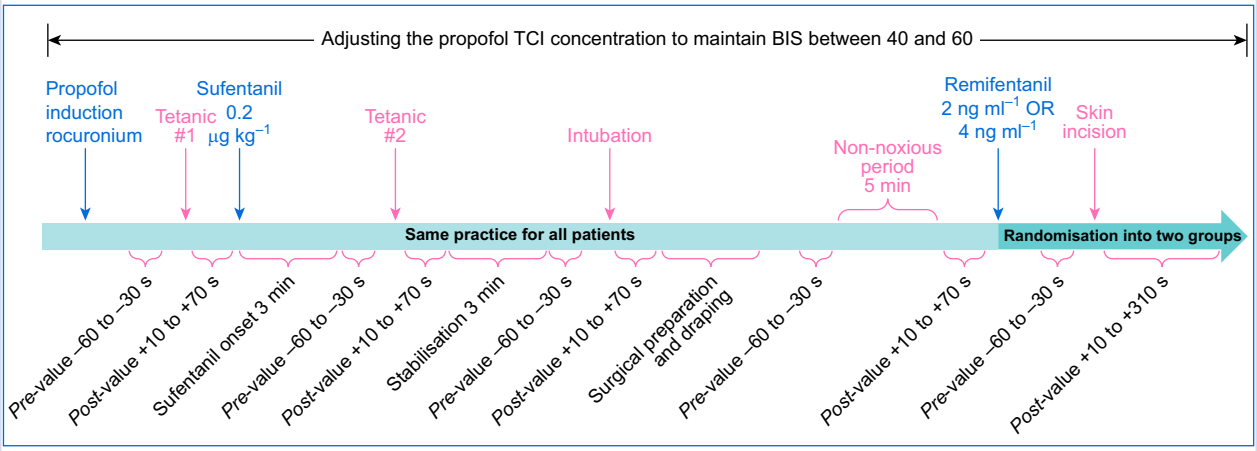


Fig 1. Study protocol. BIS, bispectral index; TCI, target-controlled infusion.

anaesthesiologist deemed the participant to be in a state of inadequate analgesia.

Data collection and study endpoints

Baseline data included participant characteristics, the American Society of Anesthesiologists (ASA) physical status classification, and surgical diagnosis. The time points of the two tetanic stimuli, intubation, skin incision, and the start and end of the non-noxious period were recorded. HR, mean arterial pressure (MAP) and BIS data were captured every 10 s and were obtained from the anaesthesia information system; Qnox and Qcon data were recorded at a 1-s interval and were exported from the Angel-6000D Multi-parameter Anaesthesia Monitor. The clocks of the anaesthesia information system and the Angel-6000D Multi-parameter Anaesthesia Monitor were accurately calibrated before the study.

Primary endpoints were the ability of the Qnox index to discriminate noxious stimuli from non-noxious stimuli and to respond to noxious stimuli. Secondary endpoints were the abilities of the Qnox index to respond to graded stimuli and discriminate different levels of analgesia.

Sample size

The sample size was estimated to detect responses to noxious stimuli and discriminate noxious stimuli from the non-noxious period. For the purpose of responding to noxious stimuli, previous results showed that the standard deviation (SD) of the Qnox was ~10.[5,10] We considered a change of 5 units as a significant difference to detect a response to nociceptive stimuli. According to pairwise multiple comparisons, aiming at a power of 0.9 and a significance level of 0.05, detecting at least one pair from among those that are actually different, the necessary sample size is 86. For the purpose of discriminating noxious stimuli from the non-noxious period, a sample size of 86 would have sufficient power to achieve the preset goal of an area under the curve (AUC) >0.70 for discrimination between noxious and non-noxious stimuli. After considering a drop-out rate of about 15%, we planned to enrol 100 participants. The sample size was calculated using PASS 11.0 software (StataCorp. LP, College Station, TX, USA).

Statistical analysis

Statistical analysis was performed using IBM SPSS Statistics for Windows version 29.0 (IBM Corp., Armonk, NY, USA) and Prism for Windows version 10.3.1 (GraphPad Software, Inc., San Diego, CA, USA). Continuous data are presented as mean (SD) in the case of a normal distribution and median (interquartile range) in the case of a non-normal distribution. Categorical data are presented as counts and percentages.

Pre-stimulus value was defined as the average value (−60 s) to (−30 s) before stimulus. We left a 30-s window before the stimulus to avoid part of the stimulus being included. Post-stimulus value was defined as average value (+10 s) to (+70 s) after tetanic 1, tetanic 2, or tracheal intubation and average value (+10 s) to (+310 s) after skin incision. We set a 10-s delay for data recording to guarantee that part of the stimulus was not included in the post-stimulus window. We determined a (+10) to (+70) s window according to a previous study that found the peak value of Qnox after stimulus occurred at ~45 s.[11] For initial incision or trocar insertion (skin incision), trocars are inserted sequentially and cannot be considered as an independent stimulus.[12] We previously observed 15 patients with thoracoscopic and laparoscopic surgery; the time to complete all trocar placements lasted an average of 240 s; therefore, we designated a (+10) to (+310) s window for analysis after skin incision. The reaction (Δ) for Qnox, Qcon, and BIS was calculated as a simple difference as follows: Average value (post-stimulus)-Average value (pre-stimulus). For HR and MAP, reaction (Δ) was normalised as: $\frac{Average\ value\ (poststimulus) - Average\ value\ (prestimulus)}{Average\ value\ (prestimulus)} \times 100$.

For the primary endpoints, in order to determine whether Qnox is capable of responding to stimuli of varying intensities, the pre-stimulus values and post-stimulus values of tracheal intubation, skin incision, tetanic 1, tetanic 2, and non-noxious period were compared by paired *t*-test, expecting a significant difference in four stimuli and no difference in the non-noxious period. Bonferroni correction was applied for five comparisons; hence, a *P*-value <0.05/5=0.01 was considered as the threshold for statistical significance. The ability to discriminate between four noxious (reaction and post-stimulus values) stimuli and the non-noxious (reaction and post-stimulus values) period was evaluated by receiver operating

characteristic (ROC) curve analysis. The optimal Youden index was calculated, and its corresponding sensitivity, specificity and cutoff value were assessed.

For the one secondary endpoint, the ability of Qnox to change proportionately to stimulus intensity was tested by a linear mixed-effects model. For post-values of Qnox, we expected the relation to be post-intubation > post-skin incision > post-tetanic 1 > post-tetanic 2 > post-non-noxious period. For the reaction value of Qnox, we expected the relation to be Δintubation > Δskin incision > Δtetanic 1 > Δtetanic 2 > Δnon-noxious period. The response to sufentanil administration was tested by comparing pre-tetanic 1 to pre-tetanic 2 values by paired *t*-test, expecting pre-tetanic 1 > pre-tetanic 2. The abilities to discriminate different levels of analgesia during noxious periods by post-skin incision values and Δskin incision were compared using *t*-test and Mann–Whitney *U*-test, respectively. Given that there are many comparisons for the secondary endpoints, an overall *P*-value of <0.001 was considered statistically significant.

In the process of the study, we found that the Qnox value changed with the Qcon value. The inventors noted that Qnox was not totally independent of the depth of anaesthesia parameter Qcon.[3] Therefore, a *post hoc* analysis was performed to explore the correlation between the two parameters. We investigated the Pearson correlation coefficient in three clinical scenarios: non-stimulus (correlation between Qnox and Qcon values in pre-tetanic 1, pre-tetanic 2, pre-intubation, pre-skin incision, pre-non-noxious period, and post-non-noxious period), mild and mild-to-moderate stimulus (correlation between Qnox and Qcon values in post-tetanic 1 and post-tetanic 2), and moderate and moderate-to-severe stimulus (correlation between Qnox and Qcon values in the post-intubation and post-skin incision periods).

Results

Participants and characteristics

A total of 166 participants were screened for eligibility, and 100 consenting patients were enrolled from May 21, 2021 to October 25, 2022. Among them, 17 participants were excluded for the following reasons: 11 for invalid measurements, 4 for protocol deviations, and 2 for cancelled surgery; therefore, 83 participants remained in the final analysis (Supplementary Fig. S1), with 42 randomised to the remifentanil TCI 2 ng ml^{-1} group and 41 to the remifentanil TCI 4 ng ml^{-1} group before skin incision. Among participants who completed the study, the mean (range) age was 57 (19–85) yr; 42 (51%) were male, and 32 underwent orthopaedic surgery, 30 thoracic surgery, 9 neurological surgery, 9 general surgery, 2 urological surgery, and 1 plastic surgery. The demographic characteristics were similar between the two groups (Table 1).

Hypnotic level and rescue sufentanil use

Our goal was target BIS values of 40–60 during the research period. Among 5727 recorded values for BIS analysis, 90.6% were within this range (419 values <40 and 122 values >60), suggesting that the hypnotic level remained stable. BIS values and Qcon values around four stimuli and the non-noxious period are presented in Table 2. Six participants received a rescue sufentanil bolus of 5 μg after tracheal intubation.

Ability of Qnox to respond to and discriminate noxious stimuli

Pre-stimulus values, post-stimulus values, and reaction after tracheal intubation, skin incision, and the tetanic 1, tetanic 2, and non-noxious periods are presented in Table 2. The post-values of Qnox increased after intubation and skin incision, reaching 53 (*P*<0.001) and 51 (*P*<0.001) on a 1–100 scale, respectively; however, statistical differences were not observed for tetanic 1 and tetanic 2. The measurements of Qnox remained unchanged after the non-noxious period. Similar changes were found in HR and MAP measurements except for a significant increase in MAP after tetanic 1.

Discrimination between four noxious stimuli (intubation, skin incision, tetanic 1, and tetanic 2) and the non-noxious period by Qnox and HR is presented in Figure 2. The AUC values for Qnox (based on the post-stimuli value) discrimination of intubation, skin incision, tetanic 1, and tetanic 2 from the non-noxious stimulus were 0.65 (95% confidence interval [CI] 0.57–0.73), 0.67 (0.58–0.75), 0.54 (0.45–0.62), and 0.52 (0.43–0.61), respectively; for ΔQnox (based on *reaction* value), the AUC values were 0.67 (0.59–0.75), 0.66 (0.58–0.75), 0.47 (0.38–0.56), and 0.43 (0.34–0.51), respectively. The AUC values for HR and MAP (based on the post-stimuli value) and ΔHR and ΔMAP (based on the reaction value) discrimination of intubation, skin incision, tetanic 1, and tetanic 2 from non-noxious stimulus are also illustrated in Figure 2. The optimal Youden index and its corresponding sensitivity, specificity and cutoff values are presented in Supplementary Table S1.

Table 1 Participant characteristics. ASA, American Society of Anesthesiologists; SD, standard deviation; TCI, target-controlled infusion.

Demographics	All patients	Patients randomised into low- and high-remifentanil-concentration groups	
	(*n*=83)	TCI 2.0 ng ml^{-1} (*n*=42)	TCI 4.0 ng ml^{-1} (*n*=41)
Age (yr), mean (range)	57 (19–85)	57 (19–82)	58 (30–85)
Height (cm), mean (SD)	166 (9)	166 (8)	165 (9)
Weight (kg), mean (SD)	67 (12)	65 (12)	69 (13)
Male, *n* (%)	42 (51)	20 (48)	22 (54)
ASA physical status, *n*(%)			
1	1 (1)	0 (0)	1 (2)
2	60 (72)	33 (79)	27 (66)
3	22 (27)	9 (21)	13 (32)

Table 2 Pre-stimulus, post-stimulus, and reaction values after four stimuli and during the non-noxious period. The pre-stimuli and post-stimuli values were presented as mean (standard deviation) and were compared by paired t-test. The reaction values were calculated as change in Qnox and Qcon and percent change in mean arterial blood pressure and heart rate and were presented as median (interquartile range). *P-value <0.01 (after Bonferroni correction) indicated statistical significance. BIS, bispectral index; Tetanic 1, tetanic stimulation without sufentanil analgesia; Tetanic 2, tetanic stimulation with sufentanil analgesia.

Parameters	Pre-stimulus	Post-stimulus	Reaction	P-value*
Qnox				
Non-noxious period	43 (11)	44 (11)	1 (−3 to 5)	0.225
Tetanic 2	46 (12)	46 (13)	0 (−5 to 4)	0.369
Tetanic 1	45 (11)	46 (13)	1 (−2 to 5)	0.116
Skin incision	47 (11)	51 (12)	4 (1−7)	<0.001
Low remifentanil	48 (11)	54 (12)	5 (1−8)	
High remifentanil	46 (11)	49 (12)	3 (1−7)	
Tracheal intubation	43 (13)	53 (15)	8 (2−18)	<0.001
Heart rate				
Non-noxious period	64 (9)	63 (9)	−1% (−6 to 3%)	0.069
Tetanic 2	67 (11)	69 (12)	0% (−2 to 3%)	0.064
Tetanic 1	71 (11)	71 (11)	0% (−3 to 3%)	0.424
Skin incision	60 (8)	64 (8)	4% (0−9%)	<0.001
Low remifentanil	61 (8)	65 (9)	6% (1−10%)	
High remifentanil	60 (9)	62 (8)	3% (0−8%)	
Tracheal intubation	66 (11)	77 (15)	13% (3−24%)	<0.001
Mean arterial blood pressure				
Non-noxious period	77 (10)	76 (8)	−1% (−8 to 4%)	0.170
Tetanic 2	78 (11)	80 (12)	0% (−2 to 4%)	0.018
Tetanic 1	78 (10)	81 (12)	2% (−1 to 6%)	<0.001
Skin incision	77 (10)	83 (11)	6% (3−13%)	<0.001
Low remifentanil	78 (10)	87 (10)	9% (4−14%)	
High remifentanil	76 (9)	80 (10)	4% (1−12%)	
Tracheal intubation	78 (11)	91 (15)	12% (6−22%)	<0.001
Qcon				
Non-noxious period	41 (8)	42 (9)	1 (−2 to 4)	0.215
Tetanic 2	43 (8)	42 (9)	-1 (−2 to 2)	0.027
Tetanic 1	42 (9)	43 (9)	0 (−2 to 2)	0.482
Skin incision	44 (8)	46 (9)	1 (−2 to 4)	0.004
Low remifentanil	45 (8)	47 (10)	1 (−2 to 4)	
High remifentanil	44 (8)	45 (9)	2 (−2 to 5)	
Tracheal intubation	41 (9)	45 (12)	2 (−2 to 8)	0.001
BIS				
Non-noxious period	47 (6)	46 (7)	−1 (−3 to 2)	0.204
Tetanic 2	49 (6)	50 (6)	1 (−1 to 3)	0.206
Tetanic 1	48 (6)	49 (7)	1 (−1 to 3)	0.038
Skin incision	48 (6)	48 (6)	0 (−2 to 3)	0.247
Low remifentanil	49 (7)	50 (7)	1 (−2 to 3)	
High remifentanil	47 (6)	47 (5)	0 (−2 to 3)	
Tracheal intubation	48 (6)	50 (8)	2 (−2 to 5)	0.001

Ability of Qnox to respond to graded stimuli

For post-stimulus Qnox values, no difference was found between any two adjacent graded-intensity stimuli (Fig. 3a). Reaction Qnox values successfully responded to graded stimuli as expected for Δintubation > Δskin incision ($P<0.001$) but not for Δskin incision > Δtetanic 1 > Δtetanic 2 > Δnon-noxious period (Fig. 3b). Compared with Qnox, MAP and HR measurements were able to discriminate more adjacent graded stimuli (Fig. 3c–f). Pairwise comparisons of Qnox between any two post-stimuli values and any two reaction values are detailed in Supplementary Table S2 and Supplementary Table S3.

Ability of Qnox to respond to opioid adminis tration and discriminate levels of analgesia

No statistically significant differences were observed between pre-tetanic 1 and pre-tetanic 2 in Qnox (45 [11] *vs* 46 [12], $P=0.170$) and MAP. The HR measurements decreased after opioid administration (pre-tetanic 1 71 [11] *vs* pre-tetanic 2 67 [11], $P<0.001$).

There was no significant difference in post-skin incision Qnox values (54 [12] *vs* 49 [12], $P=0.099$) or Δskin incision values (5 [1−8] *vs* 3 [1−7], $P=0.188$) between the low- and high-remifentanil-concentration groups during skin incision. There was no significant change in HR and MAP (Table 3).

Overall performance of Qnox, heart rate, and mean arterial pressure

Supplementary Table S4 summarises the performance of Qnox, HR, and MAP by evaluating study results according to predefined success criteria. Compared with HR and MAP, the Qnox index had less discrimination ability in nociception responses during general anaesthesia.

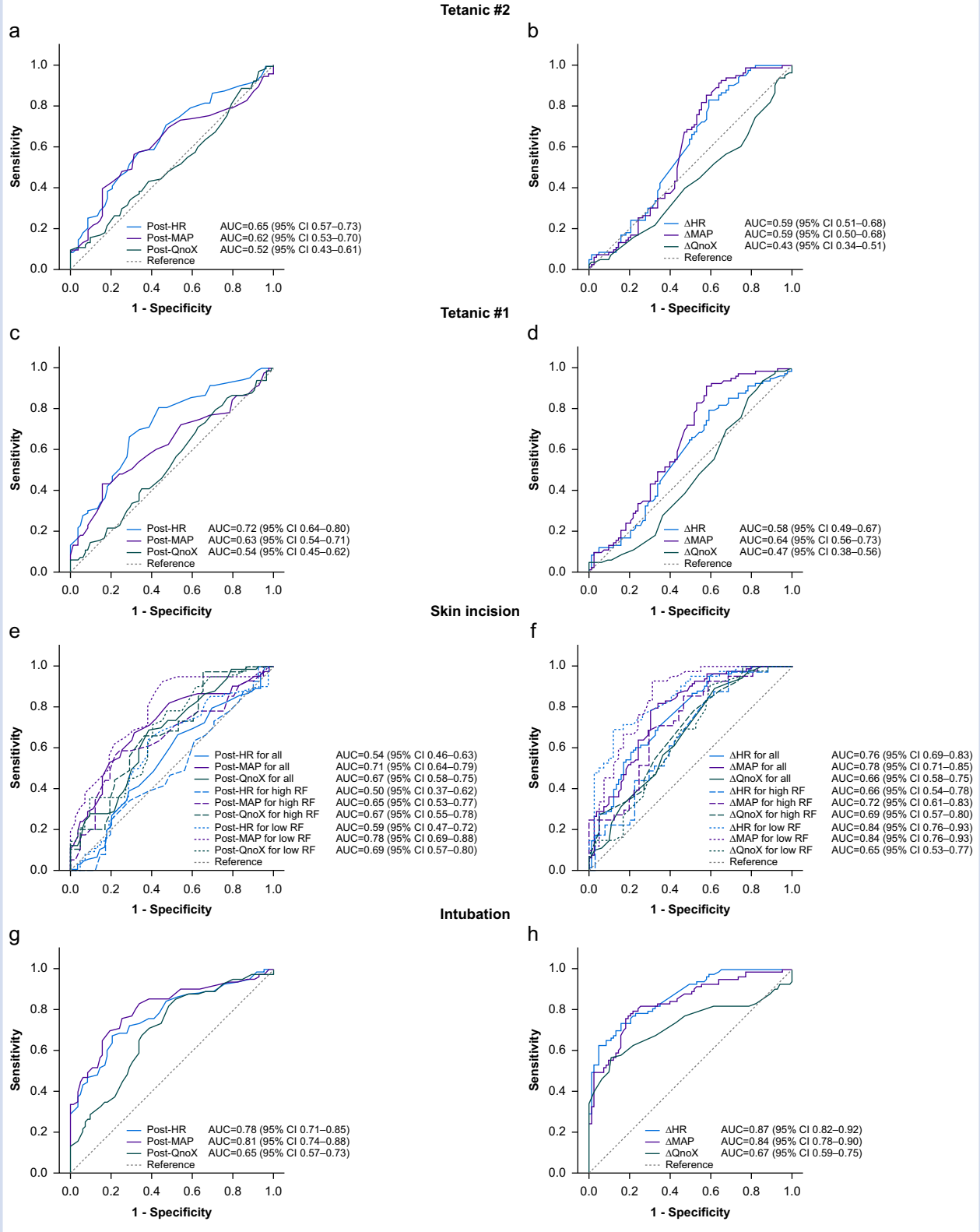


Fig 2. Receiver operating characteristic curves of post-skin incision and reaction values of Qnox, heart rate, and mean arterial pressure for discriminating noxious stimuli from the non-noxious period. Noxious stimuli: (intubation > skin incision > tetanic 1 > tetanic 2). AUC, area under the curve; HR, heart rate; MAP, mean arterial pressure; RF, remifentanil; Tetanic 1, tetanic stimulation without sufentanil analgesia; Tetanic 2, tetanic stimulation with sufentanil analgesia.

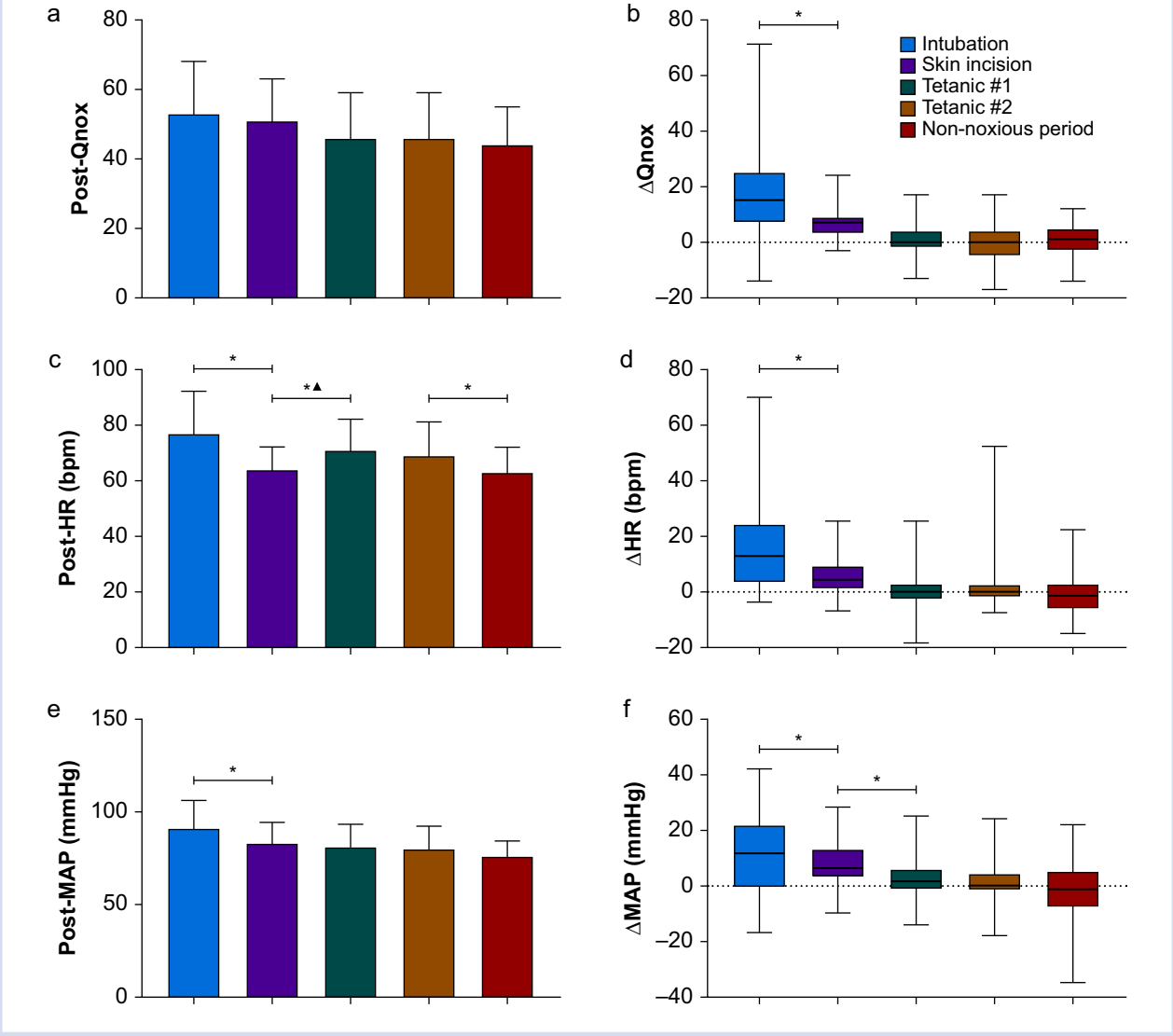


Fig 3. Graded response to four different intensities of noxious stimuli and non-noxious period by post-skin incision and reaction values. Noxious stimuli: (intubation > skin incision > tetanic 1 > tetanic 2). Tetanic 1, tetanic stimulation without sufentanil analgesia; Tetanic 2, tetanic stimulation with sufentanil analgesia. *Significant difference between the two adjacent stimuli, $P<0.001$. ▲Statistical significance in the opposite direction.

Correlation between Qnox and Qcon values

For all observed time points, the overall correlation coefficient between Qnox and Qcon was 0.830 ($P<0.001$). The Pearson correlation coefficient was 0.841 ($P<0.001$), 0.849 ($P<0.001$), and 0.791 ($P<0.001$) for non-stimuli (Supplementary Fig. S2a), mild and mild-to-moderate stimuli (Supplementary Fig. S2b), and moderate and moderate-to-severe stimuli (Supplementary Fig. S2c), respectively.

Discussion

This study provides the initial validation of the Qnox score as a measure for monitoring nociception and analgesia in patients under propofol anaesthesia with neuromuscular block. Qnox successfully responded to tracheal intubation and skin incision. However, the median ΔQnox was only 4 for skin incision and 8 for intubation. Although a statistically significant difference was achieved, whether this difference has clinical significance is unclear. Two tetanic stimulations did not result in significant changes in Qnox. Recent research can explain the results. A pilot study found it was not possible to extract any identifiable response in the EEG after standardised tetanic stimulation during routine general anaesthesia.[13] This finding provides a basis for the unchanged Qnox value after tetanic stimulation given the index was developed exclusively from EEG data. The ROC analysis was also used to evaluate the ability to discriminate between noxious stimuli and the non-noxious period. The discriminatory value of ROC curves was interpreted as excellent (AUC 0.9–1), good (0.8–0.89), fair (0.7–0.79), poor (0.6–0.69), or fail/no discriminatory capacity

Table 3 Ability to discriminate different levels of analgesia during skin incision. The pre-and post-skin incision values were presented as mean (standard deviation) and were compared by *t*-test between the two groups. *The* reaction values were calculated as change in Qnox, Qcon and mean blood pressure or percent change in heart rate and were presented as median (interquartile range) and compared by Mann–Whitney *U*-test between the two groups. *$P<0.001$ indicates statistical significance.

		Low remifentanil group (2 ng ml^{-1}) (n=42)	High remifentanil group (4 ng ml^{-1}) (n=41)	*P*-value*
Qnox	Pre-skin incision	48 (11)	46 (11)	0.246
	Post-skin incision	54 (12)	49 (12)	0.099
	Reaction	5 (1–8)	3 (1–7)	0.188
Heart rate	Pre-skin incision	61 (8)	60 (9)	0.453
	Post-skin incision	65 (9)	62 (8)	0.171
	Reaction	6% (1–10%)	3% (0–8%)	0.199
Mean arterial pressure	Pre-skin incision	78 (10)	76 (9)	0.313
	Post-skin incision	87 (10)	80 (10)	0.005
	Reaction	9% (4–14%)	4% (1–12%)	0.004

(0.5–0.59).[14] We showed that Qnox had poor predictive power for nociceptive stimuli, even weaker than HR and MAP. These results indicate that Qnox is not effective for intraoperative nociception monitoring.

An excellent tool for monitoring intraoperative nociception should be able to progressively respond to graded stimuli and discriminate different states of analgesia.[12] Our findings indicated that Qnox does not possess these capabilities. No previous study has investigated if Qnox exhibits different responses to stimuli of varying intensities. With respect to differentiating analgesic states, the results of previous studies are mixed. In one study, patients were randomised to sufentanil TCI at 0.25 ng ml^{-1} or 0.75 ng ml^{-1} during adult cardiac surgery; there were no differences in Qnox values after intubation, sternotomy, or 15 min after sternotomy, signifying that Qnox index cannot distinguish different states of analgesia during intraoperative nociceptive stimuli.[8] On the other hand, Liu and colleagues[10] randomly assigned patients to remifentanil TCI of 2 ng ml^{-1}, 4 ng ml^{-1}, or 6 ng ml^{-1} during gastroscopic polypectomy; Qnox scores were markedly higher in the first group compared with the latter two groups. However, patients did not receive neuromuscular blocking agents and significant differences in anaesthesia depth among groups might have influenced the Qnox values.

Qnox can be influenced by EMG activity and is likely to be affected by neuromuscular blocking agents.[3] Lichtenfeld and colleagues[15] showed that EMG activity significantly influenced the neuromonitoring systems because EEG and EMG signals shared an overlapping frequency range, and separating them in the clinical setting is challenging. Our results indicated small changes in Qnox after tracheal intubation (mean difference =8 [95% CI 2–18]) when compared with studies without neuromuscular blocking agents[3]; this implies that EMG signals interfere with or affect the calculation of the Qnox index. Rogozov and collegaues[8] also found a positive correlation between Qnox and EMG index after tracheal intubation but not later after the EMG index dropped, further confirming interference of EMG with Qnox.

Commonly used commercial tools for monitoring the nociception–antinociception balance, such as the surgical plethysmographic index,[16] analgesia nociception index,[17] CARDEAN index,[18] pupillometry,[19] and skin conductance,[20] are all based on changes in sympathetic response from the central nervous system during nociception or drug-induced antinociception. The Qnox index does not use autonomic tone for nociception monitoring, which avoids influence by confounding factors. However, there is a debate regarding the feasibility and reliability of a purely EEG-based tool for assessing nociception.[4] Extracting specific electrical activity triggered by pain and influenced by analgesic agents from the EEG has been challenged as futile, as most of the nociceptive effects are in the periphery, the medullary cord or deeper cerebral layers, far from the EEG signals derived from the frontal cortex.[4] During general anaesthesia, spectral changes that respond to noxious stimulation were masked by the alpha-dominant EEG rhythm induced by propofol.[13] Jensen and colleagues[3] pointed out that it is probably not possible to separately measure hypnotic (Qcon) and nociceptive effects (Qnox). Our results demonstrated significant correlations between Qnox and Qcon at the time points of both stimuli and non-stimuli, indicating that Qnox was influenced by Qcon. As Ledowski and colleagues[21] concluded, Qnox might not provide an assessment of nociceptive level *per se* but is more likely a secondary measurement of anaesthesia depth. Further research is required to elucidate how EEG signals might be used in a sensitive and specific way to reflect nociception.[22] Whether EEG-derived intraoperative nociception monitors have clinical utility is worth further exploration.

Some limitations to this study must be considered. Firstly, the recording intervals of HR or MAP (10 s) and Qnox (1 s) might impact comparison between the two parameters. Secondly, the post-stimulus value was defined as an average value of 60 s (+10 s to +70 s) or 300 s (+10 s to +310 s) after stimuli instead of peak values or averages around the peak values, which might underestimate the responsive ability of the index. Thirdly, 15 patients had hypotension (systolic arterial pressure <90 mm Hg) during the non-noxious period, and a bolus of 6 mg ephedrine i.v. was administered. This might have affected the HR. To eliminate that influence, acquisition of post-non-noxious period parameters was postponed at least 3 min after the last administration of ephedrine.

In summary, under propofol anaesthesia (BIS between 40 and 60) with neuromuscular blocking agents, the Qnox index had a limited ability to discriminate noxious stimuli from a non-noxious period. Qnox exhibited responsiveness to moderate and moderate-to-severe noxious stimuli, but not mild or mild-to-moderate stimuli. The index failed to respond to graded stimuli and was incapable of discriminating different levels of analgesia. Compared with HR and MAP, Qnox did not demonstrate greater responsiveness to intraoperative nociceptive stimuli. Further research is required to confirm or refute the utility of Qnox as a reliable tool in clinical practice.

Authors' contributions

Study conceptualisation and design: HK, HZ, D-XW
Performed the research: HK, D-DM, Y-XZ
Data collection: HK, D-DM, Y-XZ
Data entry, inspection, management: HK, D-DM, J-HM
Data analysis: HK, Y-X Z, J-HM
Data interpretation: all authors
Drafting of manuscript: HK, HZ
Critical revision of manuscript: HZ, D-XW
Final approval of manuscript: all authors

Acknowledgements

The authors gratefully acknowledge Chang-Qing Liu (Department of Anaesthesiology, Peking University First Hospital, Beijing, China) for his support in conducting the study and Xue-Ying Li (Department of Statistics, Peking University First Hospital, Beijing, China) for her help with statistical analysis.

Declarations of interest

HZ received grants from the Wu Jieping Medical Foundation (number: 320.6750.2020-21-14); HK received grants from the National High-Level Hospital Clinical Research Funding (Interdepartmental Research Project of Peking University First Hospital) (number: 2023IR22), and the Youth Clinical Research Project of Peking University First Hospital (number: 2021CR19).

Funding

Wu Jieping Medical Foundation (number: 320.6750.2020-21-14), National High-Level Hospital Clinical Research Funding (Interdepartmental Research Project of Peking University First Hospital) (number: 2023IR22), and the Youth Clinical Research Project of Peking University First Hospital (number: 2021CR19).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bja.2024.10.051.

References

1. Ledowski T. Objective monitoring of nociception: a review of current commercial solutions. *Br J Anaesth* 2019; **123**: e312–21
2. Shanthanna H, Uppal V, Joshi GP. Intraoperative nociception monitoring. *Anesthesiol Clin* 2021; **39**: 493–506
3. Jensen EW, Valencia JF, López A, et al. Monitoring hypnotic effect and nociception with two EEG-derived indices, qCON and Qnox, during general anaesthesia. *Acta Anaesthesiol Scand* 2014; **58**: 933–41
4. Raeder J. EEG-based monitor on anti-nociception during general anaesthesia: mission impossible? *Acta Anaesthesiol Scand* 2014; **58**: 911–3
5. Pantalacci T, Allaouchiche B, Boselli E. Relationship between ANI and qNOX and between MAC and qCON during outpatient laparoscopic cholecystectomy using remifentanil and desflurane without muscle relaxants: a prospective observational preliminary study. *J Clin Monit Comput* 2023; **37**: 83–91
6. Wang X, Zhang J, Feng K, et al. The effect of hypothermia during cardiopulmonary bypass on three electroencephalographic indices assessing analgesia and hypnosis during anesthesia: consciousness index, nociception index, and bispectral index. *Perfusion* 2020; **35**: 154–62
7. Melia U, Gabarron E, Agustí M, et al. Comparison of the qCON and Qnox indices for the assessment of unconsciousness level and noxious stimulation response during surgery. *J Clin Monit Comput* 2016; **31**: 1273–81
8. Rogozov V, Vanek T, Maly M, Rocek J, Kacer P. The nociceptive response during adult cardiac surgery measured by the qNOX index: a feasibility study. *Biomed Pap Med Fac Univ Palacky Olomouc Czech Repub* 2024; **168**: 62–7
9. Huiku M, Uutela K, van Gils M, et al. Assessment of surgical stress during general anaesthesia. *Br J Anaesth* 2007; **98**: 447–55
10. Liu M, Wu H, Yang D, et al. Effects of small-dose remifentanil combined with index of consciousness monitoring on gastroscopic polypectomy: a prospective, randomized, single-blinded trial. *Trials* 2018; **19**: 392
11. Boselli E, Fatah A, Ledochowski S, Allaouchiche B. Variations of qCON and qNOX during tracheal suction in ICU patients on sedation and curarization for SARS-CoV2 pneumonia: a retrospective study. *J Clin Monit Comput* 2023; **37**: 1119–21
12. Edry R, Recea V, Dikust Y, Sessler DI. Preliminary intraoperative validation of the nociception level index: a noninvasive nociception monitor. *Anesthesiology* 2016; **125**: 193–203
13. Anders M, Anders B, Dreismickenbecker E, et al. EEG responses to standardised noxious stimulation during clinical anaesthesia: a pilot study. *BJA Open* 2022; **5**, 100118
14. Fischer JE, Bachmann LM, Jaeschke R. A readers' guide to the interpretation of diagnostic test properties: clinical example of sepsis. *Intensive Care Med* 2003; **29**: 1043–51
15. Lichtenfeld F, Kratzer S, Hinzmann D, García PS, Schneider G, Kreuzer M. The influence of electromyographic on electroencephalogram-based monitoring: putting the forearm on the forehead. *Anesth Analg* 2024; **138**: 1285–94
16. Bonhomme V, Uutela K, Hans G, et al. Comparison of the surgical Pleth Index™ with haemodynamic variables to assess nociception-anti-nociception balance during general anaesthesia. *Br J Anaesth* 2011; **106**: 101–11
17. Ledowski T, Averhoff L, Tiong WS, Lee C. Analgesia Nociception Index (ANI) to predict intraoperative haemodynamic changes: results of a pilot investigation. *Acta Anaesthesiol Scand* 2014; **58**: 74–9
18. Cividjian A, Martinez JY, Combourieu E, et al. Beat-by-beat cardiovascular index to predict unexpected intraoperative

movement in anesthetized unparalyzed patients: a retrospective analysis. *J Clin Monit Comput* 2007; **21**: 91–101
19. Isnardon S, Vinclair M, Genty C, Hebrard A, Albaladejo P, Payen JF. Pupillometry to detect pain response during general anaesthesia following unilateral popliteal sciatic nerve block: a prospective, observational study. *Eur J Anaesthesiol* 2013; **30**: 429–34
20. Gjerstad AC, Storm H, Hagen R, Huiku M, Qvigstad E, Raeder J. Comparison of skin conductance with entropy during intubation, tetanic stimulation and emergence from general anaesthesia. *Acta Anaesthesiol Scand* 2007; **51**: 8–15
21. Ledowski T, Schmitz-Rode I. Predicting acute post-operative pain by the Qnox score at the end of surgery: a prospective observational study. *Br J Anaesth* 2020; **124**: 222–6
22. García PS, Kreuzer M, Hight D, Sleigh JW. Effects of noxious stimulation on the electroencephalogram during general anaesthesia: a narrative review and approach to analgesic titration. *Br J Anaesth* 2021; **126**: 445–57

Handling Editor: Hugh C Hemmings Jr